

1. Containerize a sample application using Docker

Containerize a sample application using Docker: write a Dockerfile, build the image, and run the container. Document the image layers and container behavior.

```
(kali㉿kali)~]
$ sudo systemctl start docker

(kali㉿kali)-[~]
$ sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker

(kali㉿kali)-[~]
$ docker --version
Docker version 28.5.2+dfsg4, build 9cc6dea35e9a963f281434761c656fba4ac43aed

(kali㉿kali)-[~]
$ mkdir docker-sample
cd docker-sample

(kali㉿kali)-[~/docker-sample]
$ pwd
/home/sanchi/docker-sample

(kali㉿kali)-[~/docker-sample]
$ nano app.py

GNU nano 8.0 app.py
print("Hello! This application is running inside a Docker container.")

(kali㉿kali)-[~/docker-sample]
$ cat app.py
print("Hello! This application is running inside a Docker container.")

(kali㉿kali)-[~/docker-sample]
$ nano Dockerfile
FROM python:3.12-slim

WORKDIR /app

COPY app.py .

CMD ["python", "app.py"]

(kali㉿kali)-[~/docker-sample]
$ cat Dockerfile
FROM python:3.12-slim

WORKDIR /app

COPY app.py .

CMD ["python", "app.py"]

$ sudo docker build -t sample-app .
[+] Building 35.9s (8/8) FINISHED
=> [internal] load build definition from Dockerfile 0.0s
=> [internal] load dockerfile: 115B 0.0s
=> [internal] load metadata for docker.io/library/python:3.12-slim 7.2s
=> [internal] load .dockerignore 0.0s
=> [internal] load context: 2B 0.0s
=> [1/3] FROM docker.io/library/python:3.12-slim@sha256:78387bc3881b8273120a12ebe6c1ab22b018ccc2c9adf565ae1ac9b 28.1s
=> resolve docker.io/library/python:3.12-slim@sha256:78387bc3881b8273120a12ebe6c1ab22b018ccc2c9adf565ae1ac9b5 0.0s
=> sha256:78387bc3881b8273120a12ebe6c1ab22b018ccc2c9adf565ae1ac9b536e184ea 10.37kB / 10.37kB 0.0s
=> sha256:2fe5997d249a808b8eeea52c58a1dbffbbba28754dc11699ef5c029f2d818ce79 1.75kB / 1.75kB 0.0s
=> sha256:ec7d6c95cd3692a2e2d228a8b1ca74e4025b54121fcc4c5da6f09cfa473315ad 5.66kB / 5.66kB 0.0s
=> sha256:6310eb16bf4251731feab01e8f633bf5e2d75a657ccad97f420b1f83cce457be 29.79MB / 29.79MB 19.3s
```

```

└─$ sudo docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
sample-app    latest    a498c6aaf0c1  13 seconds ago 127MB

(kali㉿kali)~/docker-sample]
└─$ sudo docker run --name sample-container sample-app
Hello! This application is running inside a Docker container.

(kali㉿kali)~/docker-sample]
└─$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
2de8b0c6b0b3  sample-app    "python app.py"         19 seconds ago Exited (0) 19 seconds ago          sample-container

(kali㉿kali)~/docker-sample]
└─$ sudo docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
2de8b0c6b0b3  sample-app    "python app.py"         19 seconds ago Exited (0) 19 seconds ago          sample-container

```

```

└─$ sudo docker history sample-app
IMAGE          CREATED          CREATED BY                                      SIZE      COMMENT
a498c6aaf0c1   About a minute ago  CMD ["python" "app.py"]                      0B        buildkit.dockerfile.v0
<missing>      About a minute ago  COPY app.py . # buildkit                      71B       buildkit.dockerfile.v0
<missing>      About a minute ago  WORKDIR /app                                  0B        buildkit.dockerfile.v0
<missing>      6 days ago        CMD ["python3"]                               0B        buildkit.dockerfile.v0
<missing>      6 days ago        RUN /bin/sh -c set -eux; for src in idle3 p... 36B       buildkit.dockerfile.v0
<missing>      6 days ago        RUN /bin/sh -c set -eux; savedAptMark="$(a... 36.8MB    buildkit.dockerfile.v0
<missing>      6 days ago        ENV PYTHON_SHA256=5c8462af5790baf43a321a1559... 0B        buildkit.dockerfile.v0
<missing>      6 days ago        ENV PYTHON_VERSION=3.12.14                    0B        buildkit.dockerfile.v0
<missing>      6 days ago        ENV GPG_KEY=7169605F62C751356D054A26A821E680... 0B        buildkit.dockerfile.v0
<missing>      6 days ago        RUN /bin/sh -c set -eux; apt-get update; a... 12MB      buildkit.dockerfile.v0
<missing>      6 days ago        ENV LANG=C.UTF-8                              0B        buildkit.dockerfile.v0
<missing>      6 days ago        ENV PATH=/usr/local/bin:/usr/local/sbin:/usr... 0B        buildkit.dockerfile.v0
<missing>      2 weeks ago       # debian.sh --arch 'amd64' out/ 'trixie' '@1... 78.6MB    debuerreotype 0.17

```